

Andrea MAIANI

Quantum-device theory, simulation, and research software

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Theoretical physicist working on quantum devices: superconducting qubits, hybrid superconductor-semiconductor devices, and mesoscopic and magnetic materials. I combine analytical theory, numerical simulation, and close collaboration with experimental teams, and I build the software this requires, from GPU solvers to physics-informed pipelines that infer device parameters from measurement data.

EXPERIENCE

January 2024 – Current	Postdoctoral Fellow (WINQ), NORDIC INSTITUTE FOR THEORETICAL PHYSICS, NORDITA Stockholm ➤ Superconducting-qubit metrology and Hamiltonian learning from measurement data: identification of higher-harmonic Josephson potentials and parameter estimation from spectroscopy and driven response. ➤ Theory of unconventional Josephson junctions, focusing on Andreev physics and higher-harmonic effects. ➤ Quantum-dot platforms for minimal Kitaev chains: realizations and identification of topological phases. ➤ Altermagnets: transport and optics in spin textures, and electrically tunable spin-qubit platforms.
August 2023 – December 2023	Postdoctoral Associate, CQIP, NEW YORK UNIVERSITY New York City ➤ Theory and modeling of Josephson-junction arrays and gate-tunable superconducting circuits. ➤ Contributed to the theory of Gatemonium, a voltage-tunable fluxonium (<i>PRX Quantum</i>, 2025). ➤ Contributed to machine-learning screening of near-surface InAs quantum-well samples.

EDUCATION

November 2019 – June 2023	Ph.D. in Theoretical Physics, UNIVERSITY OF COPENHAGEN Copenhagen ➤ Center for Quantum Devices, Niels Bohr Institute. Supervisor: Karsten Flensberg. ➤ Thesis: <i>Modeling of Hybrid Nanoelectronic Devices for Quantum Information Processing</i>. ➤ Theory of hybrid ferromagnet-superconductor-semiconductor devices and of parity-protected superconducting qubits. ➤ Electrostatic simulations of hybrid quantum devices in collaboration with Microsoft Quantum. ➤ Automated, physics-informed fitting pipeline for tunneling-spectroscopy data based on a nonlinear Usadel model, with constrained nonlinear optimization and warm-started fits across field sweeps.
September 2017 – October 2019	Double Master of Science, ALTA SCUOLA POLITECNICA Milan ➤ M.S. in Engineering Physics, Politecnico di Milano, and M.S. in Physics of Complex Systems, Politecnico di Torino; 110/110 <i>cum laude</i>. ➤ Alta Scuola Politecnica: selective joint honors program focused on innovation and entrepreneurship. ➤ Erasmus+ exchange year at KTH Royal Institute of Technology, Stockholm, with Egor Babaev. Master's thesis on vortex nucleation in superconductors, which led to two publications and the MEPGL code.
September 2014 – September 2017	Bachelor of Science in Engineering Physics, POLITECNICO DI MILANO Milan ➤ Thesis: <i>Supersymmetric and Kramers-Kronig reflectionless potentials for matter and optical waves</i>. Supervisor: Stefano Longhi. Final grade 110/110 <i>cum laude</i>.

PUBLICATIONS

15 articles in peer-reviewed journals (2020–2026), including *Physical Review Letters* (single author) and *PRX Quantum* (2), and 1 preprint; first author of 8. Google Scholar: 309 citations, h-index 10 (October 2026). Selected:

- **A. Maiani**. “Emergent Gauge Fields and Linear Dichroism in Spin-Textured Altermagnets.” *Phys. Rev. Lett.* **137**, 136703 (2026).
- J. C. Abadillo-Uriel*, **A. Maiani***, A. Cortijo, R. Aguado, and R. Seoane Souto. “All-electrical dephasing-protected spin qubits in altermagnets.” arXiv:2606.26066 (2026).
- W. M. Strickland, B. H. Elfeky, L. Baker, **A. Maiani**, J. Lee, I. Levy, J. Issokson, A. Vrajitoarea, and J. Shabani. “Gatemonium: A Voltage-Tunable Fluxonium.” *PRX Quantum* **6**, 010326 (2025).
- **A. Maiani**, A. C. C. Drachmann, L. Galletti, C. Schrade, Y. Liu, R. Seoane Souto, and S. Vaitiekėnas. “Percolative supercurrent in superconductor-ferromagnetic insulator bilayers.” *Phys. Rev. B* **111**, 174509 (2025), Editors’ Suggestion.
- **A. Maiani**, M. Kjaergaard, and C. Schrade. “Entangling Transmons with Low-Frequency Protected Superconducting Qubits.” *PRX Quantum* **3**, 030329 (2022).

* Equal contributions. Full list on Google Scholar.

SOFTWARE AND COMPUTING

Open source	MEPGL (GPU gauged string method, C++/CUDA), pyUsadel (nonlinear Usadel solver), ThermalSpin (Monte Carlo), pynakes (bibliography toolkit for LLM-assisted workflows, on PyPI), FigWorks (publication figures from code)
Quantum games	QuantumMinesweeper (2025–; FastAPI web app on a Stim/Qiskit stabilizer engine) and Qapture the Flag (2026): browser games on qubits and measurement
AI for science	Author of a mesoscopic-transport task for Terminal-Bench-Science, an open benchmark of AI agents on research workflows (2026, under final review)
Programming	Python, C++, CUDA, Mathematica
Methods	Optimization, exact diagonalization, finite-element and finite-difference methods, Monte Carlo; model identification, parameter estimation, Bayesian methods
Tools	NumPy, SciPy, pandas, JAX, PyTorch; Git, Linux, Docker, Conda, HPC clusters

FUNDING AND DISTINCTIONS

- WINQ Postdoctoral Fellowship, Nordita, 2024–present: competitive fellowship of the Wallenberg Initiative on Networks and Quantum Information.
- EDU-WACQT Summer Internship 2026 (Marianne and Marcus Wallenberg Foundation): co-applicant, SEK 160,000 for a three-month student project on superconducting qubits.
- Two Editors' Suggestions in *Physical Review B* (2025).

TALKS, TEACHING, AND SERVICE

Talks	APS March Meeting (2023), APS Global Physics Summit (2025), Gordon Research Conference on Hybrid Superconductor-Semiconductor Devices (2025), QUANTUMatter (2026); seminars at ICMM-CSIC, TU Dresden, Uppsala University, New York University, and Nordita.
Supervision	Supervised 2 Master's students and 1 Bachelor's student; co-supervised 1 further Master's student and 2 summer interns.
Teaching	Teaching assistant in three Master's courses at the University of Copenhagen (Advanced Quantum Mechanics, Condensed Matter Physics 2, Quantum Nanoelectronics; 2020–2021) and guest lecturer in Quantum Nanoelectronics (2022).
Organization	Quantum Connection Summer School (2025, 2026, 2027); Winter School on Machine Learning in Physics (2025); WINQ Workshop on Complex and Quantum Systems (2024).
Service	Panel member, AI for Natural Sciences (Stockholm University and KTH), since 2026. Reviewer for Phys. Rev. Lett., Phys. Rev. B, Phys. Rev. Applied, Phys. Rev. Materials, Nature Communications, and SciPost Physics.
Outreach	Invited workshop on quantum computing and post-quantum cryptography for Central Bank of Ireland staff (2026); popular-science talk on the 2025 Nobel Prize in Physics, Nordita (2025); ForskarFredag (2025, 2026).